Q.1 Which of the following is the representation of spherical coordinates

A. $$(r, ,z)$$

B. $$(, ,)$$

C. $$(, ,z)$$

D. $$(x, y, z)$$

Q.2 In rectangular coordinate system the point (-2,-2) is in

A. third quadrant

B. first quadrant

C. second quadrant

D. fourth quadrant

Q.3 $$Face~~ of~~ the~~ parabola~~ y = a{x^2} + bx + c ~~ is~~ opening~~ upward~~ when$$

A. a < 0

B. b < 0

C. a > 0

D. b > 0

Q.4 $${\text{The}}\,\,{\text{partial}}\,\,{\text{derivative}}\,\,{\text{with}}\,\,{\text{respect}}\,\,{\text{to}}\,\,x\,\,{\text = \frac{x}{y}\,\,{\text{is}}\,_____.$$

A. $$1$$

B. $$x$$

C. $$- {x}{{{y^2}}}$$

D. $${1}{y}$$

Q.5 $${\text{If }} ~y = x + t,~ x = {e^u} ~{\text{and}}~ t= sin u, {\text{then by chain rule, }}\frac{{dy}}{{du}} =$$

A. $${e^u} - u$$

B. $$- {e^u} - u$$

C. $$- {e^u} + u$$

D. $${e^u} + u$$

Q.6 A zero vector has magnitude equal to zero and ---------- direction.

A. Upward

B. Arbitrary

C. Downward

D. No

Q.7 Which of the following function is not the function of one variable

A. z= g(n) + 2

B. y = f(x, y)

C. m = g(s) + constant

D. y = f(x)

Q.8 If the partial derivative (w.r.t 'x' or 'y') at any point on the surface: "z=f(x,y)" is zero, then tangent plane at that point will be--------to XY-plane.

A. Perpendicular

B. Parallel

Q.9 $${\text{Value}}\,\,{\text{of}}\,\,{\text{the}}\,\,{\text{function}}\,\,{\text{ }}f(x,{\text{ }}y) = {x^3}y\,\,{\text{at}}\,\,{\text{the}}\,\,{\text{point}}\,({x^2},\,{x^3})\,\,{\text{is}}\,\,_____.$$

A. $${x^6}$$

B. $${x^9}$$

C. $${x^2}.\,{x^3}$$

D. $${x^2}.\,{x^6}$$

Q.10 $${\text{The}}\,\,{\text{partial}}\,\,{\text{derivative}}\,\,{\text{of}}\,\,f(x,y) = {x^2}{e^{ - y}}\,\,{\text{with}}\,\,{\text{respect}}\,\,{\text{to}}\,\,x\,\,{\text{is _____}}{\text{.}}$$

A. $${x^2}\,{e^{ - y}}$$

B. $$- {x^2}\,{e^{ - y}}$$

C. $${x^2}\,{e^{ - y}}$$

D. $$2x\,{e^{ - y}}$$
